

Information Geometry

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Summary

Statistical inference is constructed upon a statistical model consisting of a parameterised family of probability distributions, which forms a manifold. It is important to study the geometry of the manifold. It was Professor C. R. Rao who initiated information geometry in his monumental paper published in 1945. It not only included fundamentals of statistical inference such as the Cramér–Rao theorem and Rao–Blackwell theorem but also proposed differential geometry of a manifold of probability distributions. It is a Riemannian manifold where Fisher–Rao information plays the role of the metric tensor. It took decades for the importance of the geometrical structure to be recognised. The present article reviews the structure of the manifold of probability distributions and its applications and shows how the original idea of Professor Rao has been developed and popularised in the wide sense of statistical sciences including AI, signal processing, physical sciences and others.

Key words: information geometry; Fisher–Rao information; dual affine connections; generalised Pythagorean theorem.

1 Introduction

Information geometry studies the structure of a regular statistical model, which forms a manifold M . It consists of probability distributions parameterised by an m -dimensional vector, where the parameters constitute a coordinate system. It is a natural question to search how different two distributions in M are, that is, a distance or divergence between two distributions.

It was Professor Rao's monumental paper (Rao, 1945) that answered the question. He proposed a fundamental theory to show that M is a Riemannian manifold, where the Fisher information matrix plays the role of the Riemannian metric. At the same time, Rao presented a fundamental theorem in statistics, which was later called the Cramér–Rao theorem, because it was also independently proved by Cramér (1946). Rao calculated the Riemannian distance between two Gaussian distributions with different means and variances. The present review summarises Information Geometry initiated by C. R. Rao to show how Rao's idea has been developed.

The geometrical approach was so fundamental that it took decades for researchers to develop his idea. Chentsov (1982) followed the idea and answered the question of why the Fisher information should be used, by proposing an invariance criterion. The Fisher information matrix which Rao used is unique from the point of view of invariance under Markovian morphisms of statistical models. He further showed that a third-order symmetric tensor exists and is also unique. These two tensors together define invariant affine connections to be introduced in M .

A higher-order asymptotic theory of statistical inference was proposed by Rao (1962), where he showed that the maximum-likelihood estimator (MLE) keeps the maximum amount of information up to higher orders. When the number of observations, n , is large, it was known that the MLE is an optimal estimator, by expanding the estimation error in series of $O(1/n)$, where the ‘higher order’ implies terms of $O(1/n^2)$. By the early 1980s, higher-order asymptotic theories had been developed widely. Efron (1975) proposed a geometrical theory that the higher-order quantities are related to statistical curvatures, by developing the idea of Fisher’s unpublished research memorandum. Dawid (1975) supplemented the idea of Efron by proposing the mixture affine connection, whereas Efron used the exponential affine connection. Both belong to the class of invariant affine connections proposed by Chentsov (1982).

Following the ideas of Efron (1975) and Dawid (1975), Amari (1982) developed a geometrical theory of higher-order asymptotics, where two types of statistical curvatures play a fundamental role. The geometrical structure is the same as that of Chentsov (1982). Nagaoka and Amari (1982) further developed a dual theory of information geometry, where the two affine connections are shown to be dually coupled. In particular, the geometry of an exponential family of probability distributions is shown to be a dually flat Riemannian manifold. Such a manifold has nice geometrical properties; for instance, the generalised Pythagorean theorem and dual projection theorems hold. These properties are very useful for applications to various problems, not only in statistics but in machine learning, signal processing, game theory, physics and so on (see Amari, 2016).

The present paper celebrating Professor Rao’s 100th anniversary reviews the geometry of manifolds of probability distributions initiated by C. R. Rao and suggests some useful applications. Section 2 defines a statistical manifold and dual geometry composed on it. Section 3 focusses on a dually flat manifold based on exponential family of distributions. Section 4 touches upon estimation functions in a semiparametric statistical model. A solution is given to the Neyman–Scott problem. Section 5 deals with hierarchical statistical models. Finally, Section 6 provides conclusions.

2 Statistical Manifold and Dual Geometry

Let

$$M = \{p(x, \xi)\} \quad (1)$$

be a regular statistical model parameterised by an m -dimensional parameter vector $\xi = (\xi_1, \dots, \xi_m)$, where $p(x, \xi)$ is the probability density function of a random variable x . M is an m -dimensional manifold, where ξ is a (local) coordinate system.

The score function $s(x, \xi)$ is an m -dimensional vector whose components are defined by the derivatives of log probability,

$$s_i(x, \xi) = \partial_i \log p(x, \xi), \quad i = 1, 2, \dots, m, \quad (2)$$

where $\partial_i = \frac{\partial}{\partial \xi_i}$.

This is a fundamental quantity in statistics. For n independent observations $x_1, \dots, x_n$, the MLE $\hat{\xi}$ is the maximiser of the log likelihood and is obtained as the solution of the likelihood equations

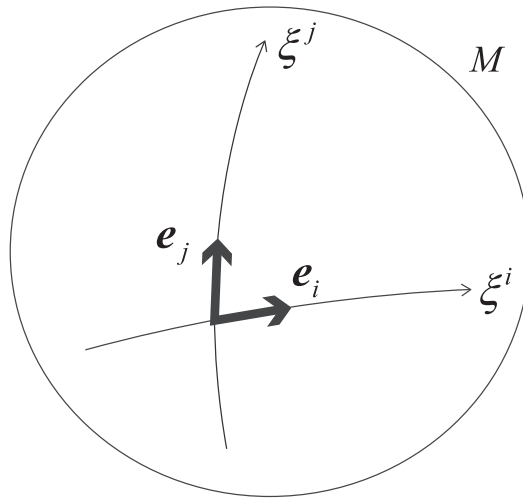


Figure 1. Coordinate curves and tangent vectors.

$$\sum_{j=1}^n s_i(x_j, \xi) = 0, \quad i = 1, \dots, m. \quad (3)$$

The score satisfies

$$E[s_i(x, \xi)] = 0, \quad (4)$$

where E is the expectation with respect to $p(x, \xi)$. The score is a vector representing how the log likelihood changes as ξ changes from ξ to $\xi + d\xi$,

$$d \log p(x, \xi) = \sum s_i(x, \xi) d\xi_i. \quad (5)$$

A component $s_i(x, \xi)$ of the score is regarded as the tangent vector e_i of M in the direction of a coordinate curve ξ_i ; see Figure 1.

This implies that the geometrical tangent vector

$$e_i = \frac{\partial}{\partial \xi^i} \quad (6)$$

is represented by the score $s_i(x, \xi)$, which is a random variable. The tangent vector $d\xi = (d\xi_i)$ is the score

$$d\xi = \sum d\xi_i e_i = \sum d\xi_i s_i(x, \xi). \quad (7)$$

When we identify a small change $d\xi$ with a small increment of the corresponding log likelihood

$$d \log p(x, \xi) = \sum d\xi_i e_i, \quad (8)$$

we have the random variable representation of tangent vectors,

$$\mathbf{e}_i = s_i(x, \xi). \quad (9)$$

We compose a tensor $g = (g_{ij})$ by

$$g_{ij}(\xi) = \langle \mathbf{e}_i, \mathbf{e}_j \rangle = E[s_i(x, \xi)s_j(x, \xi)], \quad (10)$$

where $\langle \mathbf{e}_i, \mathbf{e}_j \rangle$ is the inner product of two tangent vectors and $g = (g_{ij})$ is the Fisher information matrix. Rao (1945) used it as the Riemannian metric tensor of M so that the squared length of a small change in the parameters $d\xi$ is defined by

$$ds^2 = \sum g_{ij} d\xi_i d\xi_j, \quad (11)$$

showing that ds^2 represents the expected increment in the square of the log likelihood. We further compose a third-order symmetric tensor $T = (T_{ijk})$ in a similar way,

$$T_{ijk} = E[s_i(x, \xi)s_j(x, \xi)s_k(x, \xi)]. \quad (12)$$

Thus, a statistical manifold M is equipped with two tensors, g and T .

A statistic $t(x)$ is said to be sufficient for ξ , when the probability density of x given ξ is written in the following form:

$$p(x, \xi) = p_1(t, \xi)p_2(x), \quad (13)$$

where p_2 does not depend on ξ . Intuitively, $t(x)$ is sufficient for estimating ξ , because the latter term does not include ξ . We consider another statistical model M' ,

$$M' = \{p(t, \xi)\}, \quad (14)$$

where by an abuse of notation denoting, we let $p(t, \xi)$ be the probability density of t given ξ . Since the score vector in M' is equal to that in M , M and M' are geometrically the same. In particular, a reversible transformation of x to $y, y = f(x)$ is a sufficient statistic. This implies that the geometrical structures g and T do not depend on the representation of the random variable, x or $f(x)$.

We further require that the geometrical structure should be constructed in such a way that it is invariant under the transformation of x to its sufficient statistics. Chentsov (1982) required this by using Markov morphisms in the discrete case and proved a fundamental theorem that the invariance defines second-order and third-order symmetric tensors uniquely, which are g and T given in (10) and (12) up to a common scale.

The manifold equipped with the Fisher information Riemannian metric is not Euclidean, but curved in general. This is shown by calculating the Riemann–Christoffel curvature, where the Levi-Civita affine connection

$$\Gamma_{ijk}^0 = [ij; k] = \frac{1}{2}(\partial_i g_{jk} + \partial_j g_{ik} - \partial_k g_{ij}) \quad (15)$$

calculated from (g_{ij}) is used, as is the usual way in Riemannian geometry. However, by using T , we modify it to give two new invariant affine connections Γ and Γ^* ,

$$\Gamma_{ijk} = [ij; k] - \frac{1}{2}T_{ijk}, \quad (16)$$

$$\Gamma_{ijk}^* = [ij; k] + \frac{1}{2}T_{ijk}. \quad (17)$$

A geodesic $\xi(t)$ is a trajectory which satisfies

$$\frac{d^2}{dt^2}\xi_i(t) + \sum \Gamma_{ijk} \frac{d\xi_j}{dt} \frac{d\xi_k}{dx} = 0 \quad (18)$$

when an affine connection Γ is defined. Since there are two affine connections, we have another type of geodesic satisfying

$$\frac{d^2}{dt^2}\xi_i(t) + \sum \Gamma_{ijk}^* \frac{d\xi_j}{dt} \frac{d\xi_k}{dt} = 0. \quad (19)$$

Let

$$X = \sum X_i e_i \quad (20)$$

be a tangent vector defined at ξ . The parallel transport of a tangent vector $X = \sum X^i e_i$ along a curve

$$c: \xi_i(t) \quad (21)$$

is also defined by the vector field $X(t)$ defined on the curve by

$$\frac{d}{dt}X^i(t) + \sum_{m,j,k} \Gamma_{mjk} g^{mi} X^j \frac{d}{dt}\xi_k = 0, \quad (22)$$

where (g^{mi}) is the inverse of (g_{mi}) ; see Figure 2.

We have two parallel transports since we have two affine connections, Γ and Γ^* . They are $X(t)$ and $X^*(t)$, respectively. The inner product of vectors X and Y is given by

$$\langle X, Y \rangle = E \left[\sum X^i e_i Y^j e_j \right] = \sum X^i Y^j g_{ij}. \quad (23)$$

When the inner product of $X(t)$ and $X^*(t)$ does not change by the two parallel transports of X ,

$$\langle X(t), X^*(t) \rangle = \text{const.}, \quad (24)$$

the two affine connections are said to be dually coupled. It is straightforward to prove the following theorem (Amari & Nagaoka, 2000).

Theorem 1. *The two affine connections Γ and Γ^* are dually coupled.*

A statistical manifold is not flat in general. When it is flat, we have a local coordinate system ξ such that the affine connection vanishes

$$\Gamma_{ijk}(\xi) = 0. \quad (25)$$

A geodesic line is linear in these coordinates

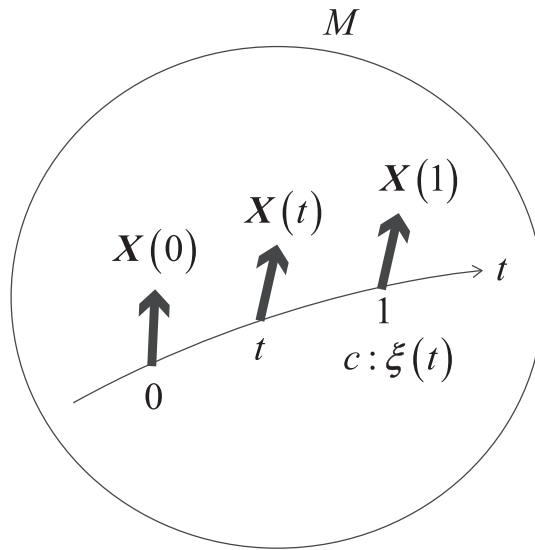


Figure 2. Parallel transport $X(t)$.

$$\xi_i(t) = c_i t + d \quad (26)$$

for constants c_i and d .

Theorem 2. For a statistical manifold M , when it is flat with respect to one affine connection, it is automatically flat with respect to the dual affine connection.

We now define a divergence and related dual structure. A function $D[p(x, \xi):p(x, \xi')]$, (or $D[\xi:\xi']$), is called a divergence, when the following criteria are satisfied:

- (1) $D[\xi:\xi'] \geq 0$,
- (2) $D[\xi:\xi'] = 0$, when and only when $\xi = \xi'$,
- (3) in the Taylor expansion $D[\xi:\xi + d\xi] = \sum g_{ij} d\xi_i d\xi_j$

for infinitesimally small $d\xi$ neglecting higher-order terms, (g_{ij}) is a positive-definite matrix.

A divergence is said to be invariant, when it is the same when we use sufficient statistic $t(x)$ instead of x . It is in general asymmetric, $D[\xi:\xi'] \neq D[\xi':\xi]$, and is regarded as the square of an (asymmetric) distance between ξ and ξ' . Rao proposed a divergence (Burbea & Rao, 1982) and studied the related Riemannian geometry. He also used a class of divergences to elucidate the performances of the M -estimators Bai *et al.* (1982).

A typical example is the Kullback–Leibler (KL-) divergence,

$$D_{KL}[\xi:\xi'] = \int p(x, \xi) \log \frac{p(x, \xi)}{p(x, \xi')} dx \quad (27)$$

and (g_{ij}) is the Fisher information matrix (hence the Riemannian metric tensor). Another example is the f -divergence (Csiszár, 1974; Morimoto, 1963) defined by using a convex function $f(u)$ satisfying

$$f(1) = 0, f'(1) = 1 \quad (28)$$

as

$$D[\xi: \xi'] = \int p(x, \xi) f \left\{ \frac{p(x, \xi')}{p(x, \xi)} \right\} dx. \quad (29)$$

It includes the α -divergence defined by

$$f_\alpha(u) = \frac{4}{1 - \alpha^2} \left(1 - u^{\frac{1+\alpha}{2}} \right). \quad (30)$$

The KL-divergence is its special case of the limit $\alpha = -1, f(u) = -\log u$.

A divergence induces a Riemannian metric together with dually coupled affine connections (Eguchi, 1983). The Riemannian metric is given by

$$g_{ij}(\xi) = \partial_i \partial_j D[\xi: \xi']|_{\xi'=\xi}. \quad (31)$$

The dual pair of affine connections are

$$\Gamma_{ijk}(\xi) = -\partial_i \partial_j \partial'_k D[\xi: \xi']|_{\xi'=\xi}, \quad (32)$$

$$\Gamma_{ijk}^*(\xi) = -\partial_k \partial'_i \partial'_j D[\xi: \xi']|_{\xi'=\xi}, \quad (33)$$

where

$$\partial'_k = \frac{\partial}{\partial \xi'_k}. \quad (34)$$

By replacing ξ and ξ' ,

$$D^*[\xi: \xi'] = D[\xi': \xi] \quad (35)$$

is called the dual divergence. It gives the same Riemannian metric and the two affine connections are interchanged. When a divergence is symmetric, it is self-dual, giving a Riemannian metric with the Levi-Civita self-dual connection.

3 Exponential Family And Dually Flat Manifold

3.1 Exponential family

An exponential family of probability distributions has the following canonical form of density:

$$p(\mathbf{x}, \boldsymbol{\theta}) = \exp\{\boldsymbol{\theta} \cdot \mathbf{x} - \psi(\boldsymbol{\theta})\} \quad (36)$$

with respect to a dominating measure $d\mu(\mathbf{x})$, where the parameter $\boldsymbol{\theta} = (\theta^1, \dots, \theta^m)$ is called the natural or canonical parameter, and $\mathbf{x} = (x_1, \dots, x_m) \in \mathbf{R}^m$ is a random variable. We use $\boldsymbol{\theta}$ having upper indices θ^i instead of ξ . A typical example is the family of Gaussian distributions,

$$p(x, \mu, \sigma) = \frac{1}{\sqrt{2\pi}\sigma} \exp\left\{-\frac{1}{2\sigma^2}(x - \mu)^2\right\}, \quad (37)$$

where the mean μ and variance σ^2 are parameters. It can be rewritten in the canonical form (36) by using

$$\mathbf{x} = (x_1, x_2) = (x, x^2), \quad (38)$$

$$\theta^1 = \frac{\mu}{\sigma^2}, \quad \theta^2 = -\frac{1}{2\sigma^2}, \quad (39)$$

and the dominating measure is defined on $x_2 - x_1^2 = 0$.

The function $\psi(\boldsymbol{\theta})$

$$\psi(\boldsymbol{\theta}) = \log \int \exp(\boldsymbol{\theta} \cdot \mathbf{x}) d\mu(\mathbf{x}), \quad (40)$$

corresponds to the normalisation constant. This is the cumulant generating function and called the free energy in physics. This is a convex function since $\partial_i \partial_j \psi(\boldsymbol{\theta})$ is positive-definite.

3.2 Geometry of an exponential family

The metric and cubic tensors are defined by (10) and (12). Calculations show that they are given by

$$g_{ij}(\boldsymbol{\theta}) = \partial_i \partial_j \psi(\boldsymbol{\theta}), \quad (41)$$

$$T_{ijk}(\boldsymbol{\theta}) = \partial_i \partial_j \partial_k \psi(\boldsymbol{\theta}). \quad (42)$$

Moreover, calculations show that the affine connection vanishes, that is,

$$\Gamma_{ijk}(\boldsymbol{\theta}) = 0 \quad (43)$$

in the $\boldsymbol{\theta}$ coordinate system. We call $\boldsymbol{\theta}$ an e -coordinate system, since it originates from the exponential family. This implies that M is flat and the affine connection vanishes in terms of $\boldsymbol{\theta}$. M is automatically dually flat. Hence, we have another coordinate system $\boldsymbol{\eta} = (\eta_1, \dots, \eta_m)$ in which the dual affine connection Γ^* vanishes. It is given by the well-known expectation parameters,

$$\eta_i = E[x_i]. \quad (44)$$

We call it an m -coordinate system, because a mixture family of probability distributions are flat with respect to mixture parameters. From (4), we have

$$\eta_i = \partial_i \psi(\boldsymbol{\theta}), \quad (45)$$

showing that $\boldsymbol{\eta}$ is obtained by the Legendre transformation (Barndorff-Nielsen, 1978). The Legendre dual of $\psi(\boldsymbol{\theta})$ is given by

$$\varphi(\boldsymbol{\eta}) = \max_{\boldsymbol{\theta}} \{\boldsymbol{\theta} \cdot \boldsymbol{\eta} - \psi(\boldsymbol{\theta})\} \quad (46)$$

and we have the identity connecting them

$$\psi(\boldsymbol{\theta}) + \varphi(\boldsymbol{\eta}) - \boldsymbol{\theta} \cdot \boldsymbol{\eta} = 0, \quad (47)$$

when $\boldsymbol{\eta}$ is given by (45).

We use the lower index for denoting the dual affine coordinates $\boldsymbol{\eta} = (\eta_i)$. The geometrical quantities are given as

$$g^{ij}(\boldsymbol{\eta}) = \partial^i \partial^j \varphi(\boldsymbol{\eta}), \quad (48)$$

$$T^{ijk}(\boldsymbol{\eta}) = \partial^i \partial^j \partial^k \varphi(\boldsymbol{\eta}), \quad (49)$$

$$\partial^i = \frac{\partial}{\partial \eta_i}, \quad (50)$$

in terms of the dual coordinate system $\boldsymbol{\eta}$, where

$$\partial^i = g^{ij} \partial_j, \quad (51)$$

and we use the Einstein summation convention that the summation symbol $\sum$ is omitted when the same index i is used in a term one as super index and the other as lower index. Therefore,

$$\partial_j = g_{ij} \partial^i \left(= \sum_i g_{ij} \partial^i \right). \quad (52)$$

Here, (g^{ij}) is the inverse of (g_{ij}) .

The KL-divergence between two distribution $p(x, \boldsymbol{\theta})$ and $p(x, \boldsymbol{\theta}')$ of exponential family is

$$D_{KL}[\boldsymbol{\theta}' : \boldsymbol{\theta}] = \psi(\boldsymbol{\theta}) - \varphi(\boldsymbol{\eta}') - \boldsymbol{\theta} \cdot \boldsymbol{\eta}', \quad (53)$$

where $\boldsymbol{\eta}'$ is the m -coordinates of $\boldsymbol{\theta}'$. We easily see that the dually flat geometrical structures are derived from the KL-divergence.

3.3 Dually flat manifold

We now review a fundamental theory of a dually flat manifold including an exponential family, but not necessarily limited to it.

Theorem 3. Fundamental theorem of dually flat manifold When M is dually flat, the following holds:

- (1) There exist two affine coordinate systems $\boldsymbol{\theta}$ and $\boldsymbol{\eta}$ with respect to two dually flat affine connections and two convex functions $\psi(\boldsymbol{\theta})$ and $\varphi(\boldsymbol{\eta})$. The dual affine connections are

$$\Gamma_{ijk}(\boldsymbol{\theta}) = 0 \quad (54)$$

in the e -coordinate system and

$$\Gamma^{*ijk}(\boldsymbol{\eta}) = 0 \quad (55)$$

in the m -coordinate system.

- (2) The Riemannian metric is given by $g_{ij}(\boldsymbol{\theta}) = \partial_i \partial_j \psi(\boldsymbol{\theta})$ in the e -coordinate system and $g^{ij}(\boldsymbol{\eta}) = \partial^i \partial^j \varphi(\boldsymbol{\eta})$ in the m -coordinate system.
- (3) The tangent vectors along the coordinate curves, given by

$$\mathbf{e}_i = \frac{\partial}{\partial \theta^i}, \quad \mathbf{e}^i = \frac{\partial}{\partial \eta^i}, \quad (56)$$

are bi-orthogonal, $\langle \mathbf{e}_i, \mathbf{e}^j \rangle = \delta_i^j$.

- (4) The cubic tensor is given by

$$T_{ijk}(\boldsymbol{\theta}) = \partial_i \partial_j \partial_k \psi(\boldsymbol{\theta}), \quad (57)$$

$$T^{ijk}(\boldsymbol{\eta}) = \partial^i \partial^j \partial^k \varphi(\boldsymbol{\eta}). \quad (58)$$

- (5) There exists a unique divergence called a canonical divergence,

$$D[\boldsymbol{\theta}' : \boldsymbol{\theta}] = \psi(\boldsymbol{\theta}) + \varphi(\boldsymbol{\eta}') - \boldsymbol{\theta} \cdot \boldsymbol{\eta}', \quad (59)$$

where $\boldsymbol{\eta}'$ denotes the m -coordinates of $\boldsymbol{\theta}'$.

The canonical divergence in a dually flat manifold of probability distributions is given by (27) and is the KL-divergence. The KL-divergence has been used in various fields such as statistics, information theory and statistical physics as a convenient measure of discrepancy between two distributions without any explicitly shown reasons. The present theory gives a justification for the KL-divergence, because it is the canonical divergence of a dually flat manifold. There are many divergences including one proposed by Rea (Burbea & Rao, 1982). They are not invariant, having the Riemannian metrics different from the Fisher information.

We remark that there are dually flat statistical manifolds other than the exponential family. A mixture family of probability distributions given by

$$M = \left\{ p(x, \boldsymbol{\eta}) = \sum_{i=0}^m \eta_i p_i(x) : \sum \eta_i = 1 \right\}, \quad (60)$$

where $p_0(x), \dots, p_m(x)$ are linearly independent prescribed probability distributions. A mixture family is not exponential family in general but is dually flat. Banerjee *et al.* (2005) showed that, under a certain regularity condition, a dually flat manifold M is related to an exponential family provided the inverse Laplace transform of $\psi(\boldsymbol{\theta})$ exists.

Given a convex function $\psi(\boldsymbol{\theta})$, we can introduce a dually flat geometrical structure in which $\boldsymbol{\theta}$ is a flat coordinate system and its dual is given by the Legendre transform

$$\boldsymbol{\eta} = \partial_i \psi(\boldsymbol{\theta}). \quad (61)$$

The canonical divergence is rewritten as

$$D[\theta':\theta] = \psi(\theta) - \psi(\theta') - \nabla\psi(\theta') \cdot (\theta - \theta'). \quad (62)$$

and its dual is

$$D^*[\theta:\theta'] = D[\theta':\theta] = \psi(\theta') - \psi(\theta) - \nabla\psi(\theta) \cdot (\theta' - \theta). \quad (63)$$

This is called the Bregman divergence derived from a convex function.

3.4 Pythagorean theorem and projection theorem

A dually flat manifold M has a nice properties. Given three points $P, Q, R \in M$, we connect P and Q by the m -geodesic and Q and R by the e -geodesic. Let $\theta_P, \theta_Q, \theta_R$ be their e -coordinates and η_P, η_Q, η_R be their m -coordinates. Then, the e -geodesic connecting Q and R is

$$\theta(t) = (1 - t)\theta_Q + t\theta_R, \quad (64)$$

and its tangent vector is given by

$$\dot{\theta} = \theta_R - \theta_Q. \quad (65)$$

The m -geodesic connecting P and Q is given by

$$\eta(t) = (1 - t)\eta_P + t\eta_Q \quad (66)$$

and its tangent vector is given by

$$\dot{\eta}(t) = \eta_Q - \eta_P. \quad (67)$$

Theorem 4. Pythagorean theorem (Figure 3). When the m -geodesic connecting P and Q is orthogonal to the e -geodesic connecting Q and R ,

$$D[P:R] = D[P:Q] + D[Q:R], \quad (68)$$

where D is the canonical divergence. Dually, when the e -geodesic connecting P and Q is orthogonal to the m -geodesic connecting Q and R

$$D^*[P:R] = D^*[P:Q] + D^*[Q:R]. \quad (69)$$

The projection theorem follows directly from the Pythagorean theorem. Let S be a smooth submanifold in a dually flat manifold M . Let P be a point outside S . We search for the minimiser $\hat{P} \in S$ of $D[P:Q]$, $Q \in S$, or dually the minimiser of $D[Q:P] = D^*[P:Q]$.

Theorem 5. Projection theorem (Figure 4). The minimiser of $D[P:Q]$, $Q \in S$ is obtained by m -projecting P to S such that m -geodesic connecting P and $\hat{P}$ is orthogonal to S .

Dually, the minimiser of $D^*[P:Q]$ is obtained by e -projecting P to S such that the e -geodesic connecting P and $\hat{P}^*$ is orthogonal to S . The m -projection is unique, when S is e -flat and the

e-projection is unique, when *S* is *m*-flat. The proof is immediate from the Pythagorean theorem. The two theorems are useful in various applications (Amari, 2016).

3.5 Statistical inference in a curved exponential family

Let $D = \{\mathbf{x}_1, \dots, \mathbf{x}_n\}$ be n independently observed data from an exponential family. The MLE $\hat{\boldsymbol{\eta}}$ is simply given by the arithmetic mean of the data,

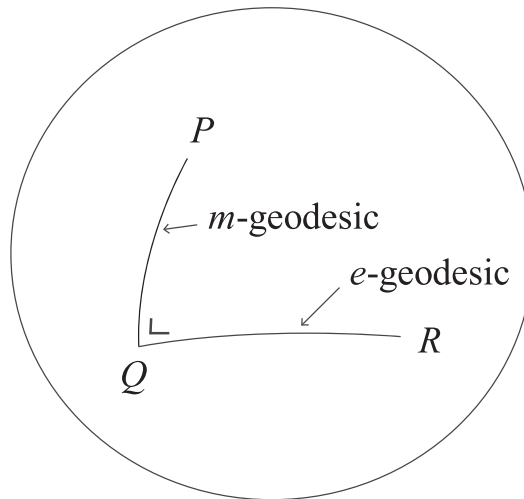


Figure 3. Pythagorean theorem.

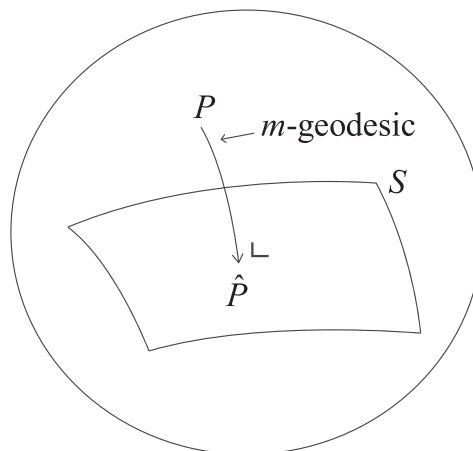


Figure 4. Projection theorem.

$$\hat{\boldsymbol{\eta}} = \frac{1}{n} \sum \mathbf{x}_i \quad (70)$$

in terms of the dual parameter $\boldsymbol{\eta}$. We call the point $\hat{\boldsymbol{\eta}} \in M$ the observed point given directly from the data D . The empirical distribution of D does not belong to M but the observed point lies in M . The observed point converges to the true point as n tends to infinity.

We now consider statistical inference in a curved exponential family $q(\mathbf{x}, \mathbf{u})$ specified by parameters $\mathbf{u} = (u^a) = (u_1, \dots, u_r)$, $r < m$. It is a submodel embedded in M , called an (m, r) -exponential family.

$$S = \{q(\mathbf{x}, \mathbf{u}) \mid q(\mathbf{x}, \mathbf{u})\} = \exp[\boldsymbol{\theta}(\mathbf{u}) \cdot \mathbf{x} - \psi\{\boldsymbol{\theta}(\mathbf{u})\}]. \quad (71)$$

Estimation is regarded as a projection of the observed point $\hat{\boldsymbol{\eta}}$ to $S \subset M$. The MLE $\hat{\mathbf{u}}$ is given by the minimiser of the log likelihood, or equivalently the KL-divergence from $\hat{\boldsymbol{\eta}}$ to S ,

$$\hat{\mathbf{u}} = \arg \min_S D_{KL}[\boldsymbol{\eta}; S]. \quad (72)$$

Because of the Pythagorean theorem, it is given by the m -projection of the observed point $\hat{\boldsymbol{\eta}}$ to S . It is Fisher efficient, and the asymptotic variance of the MLE is given by

$$\text{Cov}[\hat{\mathbf{u}}] = \frac{1}{n} B_a^i g_{ij} B_b^j \quad (73)$$

where matrix B is

$$B_a^i = \frac{\partial \theta^i(\mathbf{u})}{\partial u^a}. \quad (74)$$

The matrix

$$g_{ab} = B_a^i B_b^j g_{ij} \quad (75)$$

is the Fisher information matrix of model S .

Higher-order theory of statistical estimation was initiated by Rao (Rao, 1962; Rao *et al.*, 1982). It is possible to study higher-order asymptotics of an estimator in terms of geometry, where the e - and m -curvatures play a fundamental role (see, e.g. Amari, 1985; 2016; Amari & Nagaoka, 2000). But it is complicated, so we show only results.

Theorem 6. *An estimator is efficient when it is given by the orthogonal projection of $\hat{\boldsymbol{\eta}}$ to S . It is higher-order efficient when the orthogonal projection is m -flat. The higher-order asymptotic error covariance is decomposed in a sum of the m -curvature of the projection trajectory, the e -curvature of the model S , and a connection term.*

For higher-order asymptotic theory of statistical testing; see Kumon & Amari (1983) and Amari & Nagaoka (2000).

4 Geometry of Estimation Functions in Semiparametric Statistical Models: The Neyman–Scott Problem

4.1 Semiparametric model and nuisance parameter

A semiparametric model

$$S = \{p(\mathbf{x}, u)\} \quad (76)$$

is specified by a parameter of interest u , but the form of p is unknown, and it has infinite degrees of freedom. Here, we assume that u is a scalar, but it is easy to generalise to the vector case. A simple example is a location model,

$$S = \{p(x, u)\}, \quad (77)$$

where p is an arbitrary smooth function having moments and satisfying

$$\int p(x)dx = 1, \quad \int xp(x)dx = 0. \quad (78)$$

We study here the mixture-type semiparametric model to describe the Neyman–Scott problem. To this end, we define a regular model

$$Q = \{q(\mathbf{x}, u, \xi)\} \quad (79)$$

having two types of parameters u and ξ . Here, u is the parameter of interest and ξ is the nuisance parameter which we do not care. Let $D = \{\mathbf{x}_1, \dots, \mathbf{x}_n\}$ be n independent observations, where $\mathbf{x}_i$ is an observation from $q(\mathbf{x}, u, \xi_i)$. Here, the parameter of interest u is common for all observations but ξ_i may differ each time. This model is proposed by Neyman & Scott (1948), in which they showed that the MLE is not necessarily consistent nor efficient. Then, what is the best estimator in this situation in the asymptotic sense? The Neyman–Scott problem had bothered statisticians for many years until Bickel *et al.* (1994) and Amari & Kawanabe (1997) presented convincing theories.

It is convenient to assume that unknown values ξ_i of the nuisance parameter are randomly generated from an unknown probability distribution $k(\xi)$. Then, each $\mathbf{x}_i$ is regarded as an iid random variable generated from a distribution belonging to the model

$$S_M = \{p(\mathbf{x}, u, k)\}, \quad (80)$$

$$p\{\mathbf{x}, u, k\} = \int q(\mathbf{x}, u, \xi)k(\xi)d\xi. \quad (81)$$

S_M includes the nuisance parameter $k(\xi)$ of function degrees of freedom and is called a semiparametric model (Begun *et al.*, 1983).

S_M has function degrees of freedom, so we need to treat a function space of probability distributions. We do not present a rigorous geometrical theory, which has not yet been completed (Ay *et al.*, 2017; Pistone & Sempi, 1995). Instead, we give an intuitive arguments, without specifying the conditions under which our theory holds. But the theory is useful for practical applications.

Let us consider the tangent space of S_M at point (u, k) . Since a small deviation in log probability

$$w(\mathbf{x}) = \delta \log p(\mathbf{x}, u, k) \quad (82)$$

satisfies

$$E[w(\mathbf{x})] = 0, \quad (83)$$

we consider

$$T_{u,k} = \{w(\mathbf{x}) \mid E[w(\mathbf{x})] = 0\}, \quad (84)$$

where E is expectation with respect to $p(\mathbf{x}, u, k)$, as the tangent space of S_M at (u, k) , provided

$$E[\{w(\mathbf{x})\}^2] < \infty \quad (85)$$

is satisfied. Then the tangent vectors $w(\mathbf{x})$ form a Hilbert space. The inner product of two tangent vectors $w_1(\mathbf{x})$ and $w_2(\mathbf{x})$ is given by

$$\langle w_1(\mathbf{x}), w_2(\mathbf{x}) \rangle = E[w_1(\mathbf{x})w_2(\mathbf{x})]. \quad (86)$$

It includes a one-dimensional subspace

$$T_u = \left\{ \frac{d}{du} \log p(\mathbf{x}, u, k) \right\}, \quad (87)$$

which represents a change in the direction of u . It also includes

$$T_k = \left\{ \frac{d}{dt} \log p(\mathbf{x}, u, c(t)) \right\}, \quad (88)$$

where $k = c(t)$ is a curve in S_M passing through (u, k) at $t = 0$. This is the nuisance tangent space of infinite dimensions. The above two do not cover entire $T_{u,k}$ and it includes other vectors $a(\mathbf{x})$ which are orthogonal to both T_u and T_k . They form a subspace called auxiliary tangent subspace T_a . The tangent space is decomposed in a direct sum,

$$T_{u,k} = T_u \oplus T_k \oplus T_a, \quad (89)$$

where T_a is orthogonal to $T_u \oplus T_k$.

4.2 Estimating functions

A function $f(\mathbf{x}, u)$ is called an estimating function (Godambe, 1991), when it satisfies

$$E_{u,\xi}[f(\mathbf{x}, u)] = 0, \quad (90)$$

$$A = E_{u,\xi} \left[\frac{d}{du} f(\mathbf{x}, u) \right] > 0, \quad (91)$$

where $E_{u,\xi}$ represents expectation with respect to $q(\mathbf{x}, u, \xi)$ for any ξ . Note that (90) and (91)

hold for expectation with respect to any $p(\mathbf{x}, u, k)$, which is a linear combination of $q(\mathbf{x}, u, \xi)$ s. An estimator $\hat{u}$ is obtained from

$$\sum f(\mathbf{x}_i, u) = 0, \quad (92)$$

where the expectation in (90) is replaced by the arithmetic sum. It is easy to prove that the estimator is consistent and asymptotically Gaussian having the variance

$$V[\hat{u}] = \frac{1}{n} \frac{E[\{f(\mathbf{x}, u)\}^2]}{A^2}, \quad (93)$$

in a similar way as we prove that the MLE is asymptotically Gaussian (Lindsay, 1982).

We need to show when an estimating function exists and which is better when there are many. In order to characterise the estimation function, we use parallel transports of tangent vectors. The e - and m -transports of a tangent vector $w(\mathbf{x})$ from (u, k_1) to (u, k_2) are defined, respectively, by

$$\overset{e}{\Pi}_{k_1}^{k_2} w(\mathbf{x}) = w(\mathbf{x}) - E_{u, k_2}[w(\mathbf{x})], \quad (94)$$

$$\overset{m}{\Pi}_{k_1}^{k_2} w(\mathbf{x}) = \frac{p(\mathbf{x}, u, k_1)}{p(\mathbf{x}, u, k_2)} w(\mathbf{x}). \quad (95)$$

It is easy to see that the parallel transports keep the inner product invariant in the following sense:

$$\langle w_1(\mathbf{x}), w_2(\mathbf{x}) \rangle_{u, k_1} = \langle \overset{e}{\Pi}_{k_1}^{k_2} w_1(\mathbf{x}), \overset{m}{\Pi}_{k_1}^{k_2} w_2(\mathbf{x}) \rangle_{u, k_2}. \quad (96)$$

Theorem 7. *An estimating function $f(\mathbf{x}, u)$ is orthogonal to the nuisance tangent space,*

$$\langle f(\mathbf{x}, u), v(\mathbf{x}, u, \xi) \rangle = 0, \quad (97)$$

$$v(\mathbf{x}, u, \xi) = \frac{d}{d\xi} \log q(\mathbf{x}, u, \xi) \quad (98)$$

and is invariant under the e -parallel transport from any (u, k_1) to (u, k_2) ,

$$\overset{e}{\Pi}_{k_1}^{k_2} f(\mathbf{x}, u) = f(\mathbf{x}, u). \quad (99)$$

We omit the proof, because it is easily obtained from the definition.

Let

$$s(\mathbf{x}, u, k) = \frac{d}{du} \log p(\mathbf{x}, u, k) \quad (100)$$

be the score vector and let $s_I(\mathbf{x}, u, k)$ be its projection to the subspace orthogonal to T_k . We call it an information score. We then have the following theorem.

Theorem 8. *An estimating function is given by the following sum:*

$$f(\mathbf{x}, u) = s_I(\mathbf{x}, u) + a(\mathbf{x}), \quad a(\mathbf{x}) \in T_a. \quad (101)$$

This proves that an estimating function exists when the information score is non-vanishing. The optimal estimating function is $s_I(\mathbf{x}, u, k)$, which we do not know since k is unknown. However, this gives good means of selecting an estimating function. Even when we choose k incorrectly, it is still an estimating function. When the true nuisance function is k_0 , the optimal estimating function is $s_I(\mathbf{x}, u, k_0)$. An estimating function using the incorrect k is written as

$$s_I(\mathbf{x}, u, k) = s_I(\mathbf{x}, u, k_0) + a(\mathbf{x}), \quad (102)$$

by using some auxiliary $a(\mathbf{x})$, and it still gives a consistent estimator. This is not true when we use statistical model $p(\mathbf{x}, u, k)$ for estimating u with the incorrect k .

4.3 Estimation of rate of proportionality in a linear model

We focus on a specific example of Neyman–Scott problem, although we can analyse other problems similarly. Let u be the parameter of interest, which represents the ratio of weight to volume of sample material,

$$u = \frac{w}{v}, \quad (103)$$

where v is the volume and w is its weight of a specimen. We assume we have the following observations (x_i, y_i) , where x_i are noisy observations of volumes and y_i are noisy observations of the weights of various specimens. They are given by

$$x_i = \xi_i + \varepsilon_i, \quad (104)$$

$$y_i = u\xi_i + \varepsilon'_i, \quad (105)$$

where ε_i and ε'_i are subject to a Gaussian distribution with mean 0 and variance σ^2 independently.

The joint distribution of $\mathbf{x} = (x, y)$ is written as

$$q(x, y, u, \xi) = \frac{1}{2\pi\sigma^2} \exp\left\{-\frac{1}{2\sigma^2}[(x - \xi)^2 + (y - u\xi)^2]\right\}. \quad (106)$$

We rewrite (106) as

$$q(x, y, u, \xi) = \exp\{\xi s(x, y, u) + r(x, y) - \psi(u, \xi)\}, \quad (107)$$

where we put

$$s(x, y, u) = x + uy, \quad (108)$$

$$r(x, y) = -\frac{1}{2}(x^2 + y^2). \quad (109)$$

Since ξ is a random variable subject to $k(\xi)$ in S_M , we can study the conditional distribution of ξ given s in (106). This is given by

$$p(\xi|s) = \frac{k(\xi)\exp\{\xi s + r - \psi\}}{p(x, u, k)}. \quad (110)$$

We have

$$E[\xi|s] = h(uy + x), \quad (111)$$

where h is a function depending on k . Using this, we have an explicit form of the nuisance tangent space T_k . It is spanned by s irrespective of k ,

$$T_k = \{s(x, y, u)\}. \quad (112)$$

Hence, it is invariant under the parallel transport from k_1 to k_2 . We further calculate the information score, obtaining

$$s_I(x, y, u) = (y - ux)E[\xi|s]. \quad (113)$$

This proves that an estimation function is of the form,

$$f(x, y, u) = (y - ux)h(uy + x), \quad (114)$$

where $h(z)$ is an arbitrary function. The best function h depends on the unknown k .

There are a number of estimators in this specific problem. We show some of them in the following:

1) Least square estimator, which minimises

$$L = \frac{1}{2} \sum (u_i - ux_i)^2 \quad (115)$$

and the solution is simply given by

$$\hat{u} = \frac{\sum y_i x_i}{\sum x_i^2}. \quad (116)$$

2) MLE or the total least square

This is the estimator maximising log likelihood of all observations with respect to all the unknown parameters u and $\xi_1, \dots, \xi_n$. This minimises the sum of squares of the lengths of lines that project the observed (x_i, y_i) to the regression line

$$y = ux. \quad (117)$$

The solution is given by solving

$$\sum (y_i - ux_i)(uy_i + x_i) = 0. \quad (118)$$

3) Ratio of the sum of total weight to the sum of total volume, simply given by

$$\hat{u} = \frac{\sum y_i}{\sum x_i}. \quad (119)$$

The least square solution is not good in this case, since it does not give even a consistent estimator. Both MLE and the ratio estimators are consistent. Which is better? It depends on the distribution $k(\xi)$. Roughly speaking, when the average of $k(\xi)$ is much larger than its standard deviation, MLE is better, but when it is smaller, the ratio estimator is better.

Amari & Kawanabe (1997) proposed an estimator of which estimating function includes a parameter c ,

$$f(x, y, u) = (y - ux)(uy + x + c). \quad (120)$$

When $c = 0$, it gives MLE. When $c \rightarrow \infty$, it gives the ratio estimator. They proposed a method of determining c from observed data.

5 Hierarchical Model and Higher-Order Interactions

Many statistical models have hierarchical structures such that lower-order submodels are included in higher-order submodels. Markov chains consisting of various orders are such models in which a k -th order submodel is included in a $(k+1)$ th order submodel. An autoregression (AR) model in time series is another example, in which a k -th order submodel is included in a $(k+1)$ -th order submodel.

We study a hierarchical structure of an exponential family S_n consisting of

$$p(\mathbf{x}, \boldsymbol{\theta}) = \exp \left[\sum_{i=1}^n \boldsymbol{\theta}^{(i)} \cdot \mathbf{x}^{(i)} - \psi \left\{ \boldsymbol{\theta}^{(1)}, \dots, \boldsymbol{\theta}^{(n)} \right\} \right], \quad (121)$$

where $\mathbf{x} = (\mathbf{x}^{(1)}, \dots, \mathbf{x}^{(n)})$, $\boldsymbol{\theta} = (\boldsymbol{\theta}^{(1)}, \dots, \boldsymbol{\theta}^{(n)})$. The corresponding $\boldsymbol{\eta}$ -coordinates are $\boldsymbol{\eta} = (\boldsymbol{\eta}_{(1)}, \dots, \boldsymbol{\eta}_{(n)})$. Let S_k be a submodel defined by

$$S_k = \left\{ p(\mathbf{x}, \boldsymbol{\theta}) \mid \boldsymbol{\theta}^{(k+1)} = \dots = \boldsymbol{\theta}^{(n)} = 0 \right\}. \quad (122)$$

Then, we have a hierarchical structure $S_1 \subset S_2 \dots \subset S_n$.

We divide the $\boldsymbol{\theta}$ coordinates into two parts, $\boldsymbol{\theta} = (\boldsymbol{\theta}_k; \boldsymbol{\theta}^{(k+1)})$, $\boldsymbol{\theta}_k = (\boldsymbol{\theta}^{(1)}, \dots, \boldsymbol{\theta}^{(k)})$, $\boldsymbol{\theta}^{(k+1)} = (\boldsymbol{\theta}^{(k+1)}, \dots, \boldsymbol{\theta}^{(n)})$, which is called a k -cut. We also divide the $\boldsymbol{\eta}$ coordinates as $\boldsymbol{\eta} = (H_k; H^{(k+1)})$, $H_k = (\boldsymbol{\eta}_{(1)}, \dots, \boldsymbol{\eta}_{(k)})$, $H^{(k+1)} = (\boldsymbol{\eta}_{(k+1)}, \dots, \boldsymbol{\eta}_{(n)})$. Since $\boldsymbol{\theta}^i$ is orthogonal to

η_j ($j \neq i$), or more precisely the tangent vector $\mathbf{e}_i$ along the coordinate θ^i is orthogonal to the tangent vector $\mathbf{e}^j$ along the coordinate η_j ($j \neq i$), parameters Θ^{k+1} are orthogonal to parameters H_k .

It is convenient to define a new coordinate system, $\Xi = (H_k, \Theta^{k+1})$, which consists of the lower part of $\boldsymbol{\eta}$ coordinates and the higher part of $\boldsymbol{\theta}$ coordinates. We call it a mixed coordinate system (Amari, 2001). Let us consider a submanifold specified by

$$H_k = (\mathbf{c}^{(1)}, \dots, \mathbf{c}^{(k)}) \quad (123)$$

for fixed vectors $\mathbf{c}^{(1)}, \dots, \mathbf{c}^{(k)}$. Complementary, we consider a submanifold specified by $\Theta^{k+1} = (\mathbf{d}_{(k+1)}, \dots, \mathbf{d}_{(n)})$ for fixed vectors $\mathbf{d}_{(k+1)}, \dots, \mathbf{d}_{(n)}$. These two submanifolds are orthogonal to each other for any $\mathbf{c}$ and $\mathbf{d}$. Note that two submanifolds H_k and H^{k+1} or Θ_k and Θ^{k+1} are not orthogonal.

We consider a simple binary model for $\mathbf{x} = (x_1, \dots, x_n)$, each x_i taking 0 or 1. Given an observed contingency table $T(x_1, x_2, \dots, x_n)$, we have interest in knowing how x_i s are mutually interrelated in probability distribution $p(x_1, \dots, x_n)$. There are pairwise interactions, three-way interactions and further higher-order interactions among variables $x_1, \dots, x_n$. How can we quantify these interactions? The mixture coordinates and the k -cuts are useful for defining the degrees of interactions among random variables. We show that the interaction terms can be decomposed orthogonally into pairwise, triplewise and higher-order ones.

We use a neural network of binary neurons as a typical example for intuitive explanation. The network consists of n connected neurons interacting one another. Each neuron takes two states, excited or quiescent. Let $x_1, \dots, x_n$ be n binary random variables, where $x_i = 1$ represents that the i -th neuron is excited and $x_i = 0$ quiescent. The current state $\mathbf{x} = (x_1, \dots, x_n)$ is regarded as a vector random variable, and its joint probability distribution $p(x_1, \dots, x_n)$ is shown in the form of an exponential family,

$$p(\mathbf{x}, \boldsymbol{\theta}) = \exp \left\{ \sum \theta^i x_i + \sum \theta^{ij} x_i x_j + \dots + \theta^{12 \dots n} x_1 \dots x_n - \psi(\boldsymbol{\theta}) \right\}. \quad (124)$$

Its e -coordinates are organised as $\boldsymbol{\theta} = (\Theta^1, \Theta^2, \dots, \Theta^n)$, where subvectors $\Theta^k = \{\theta^{i_1 \dots i_k}\}$, $i_1 < \dots < i_k$, summarise the k -th order terms.

The corresponding m coordinates are $H_k = \{\eta_{i_1 \dots i_k}\}$, where $\eta_{i_1 \dots i_k} = E[x_{i_1} \dots x_{i_k}]$. $\boldsymbol{\eta}$ is decomposed as

$$\boldsymbol{\eta} = (H_1, \dots, H_n). \quad (125)$$

The subvector H_k represents the probability of k neurons jointly firing, that is, the probability of $x_{i_1} x_{i_2} \dots x_{i_k} = 1$. It is important to see that the coordinate axes of H_k are orthogonal to those of Θ^{kt} when $kt > k$. A submodel

$$S_k = \{\boldsymbol{\theta} | \Theta^{k+1} = \dots = \Theta^n = 0\} \quad (126)$$

is called a k -th order model. We have a hierarchical structure $S_0 \subset S_1 \subset \dots \subset S_n$, where S_0 is a single point specified $\boldsymbol{\theta} = 0$.

We begin with a simple model of two neurons, $n = 2$. The probability distribution of $\mathbf{x}$ is

$$p(\mathbf{x}, \boldsymbol{\theta}) = \exp\{\theta^1 x_1 + \theta^2 x_2 + \theta^{12} x_1 x_2 - \psi(\boldsymbol{\theta})\}. \quad (127)$$

When $\theta^{12} = 0$, two neurons fire independently, $p(\mathbf{x}) = p_1(x_1)p_2(x_2)$ and there are no mutual interactions between the two. When two neurons are not independent, there is mutual interaction. We want to know its degree. The covariance $\sigma^2 = E[x_1 x_2] - E[x_1]E[x_2]$ is 0, when the two are independent, which clearly shows a degree of mutual interaction. However, there are many such quantities that vanish when the two neurons fire independently. The increase in log likelihood, due to an increase in firing rates, is correlated to the increase of log likelihood due to the covariance. So the firing rates and the covariance are not separated. Geometrically speaking, we want to have such a quantity that the direction due to an increase in interaction is orthogonal to the directions due to increases in firing rates. This quantity represents the degree of interaction that does not change even when the firing rates change. The covariance is not such a quantity.

We know that the e -coordinate θ^{12} is orthogonal to firing rates η_1 and η_2 , so it represents the interaction orthogonal to the firing rates. As is seen in Figure 5, S_1 defined by $\theta^{12} = 0$ is an e -flat submanifold consisting of all the independent distributions.

Given a general distribution, we m -project it to S_1 . Then, we have a distribution in S_1 which has the same firing rates but no interaction. The submanifold M_1 defined by $\eta_1 = c_1, \eta_2 = c_2$, for constants c_1 and c_2 , is m -flat, consisting of the distributions having the same firing rates c_1 and c_2 but the degree of interaction θ^{12} is different. They are orthogonal intersecting at $\eta_i = c_i, \theta^{12} = 0$.

Let p_0 be the distribution specified by $\boldsymbol{\theta} = 0$, that is $\eta_1 = \eta_2 = 1/2$ and $\theta^{12} = 0$. The KL-divergence from p to p_0 is decomposed as

$$D_{KL}[p:p_0] = D_{KL}[p:\hat{p}] + D_{KL}[\hat{p}:p_0], \quad \hat{p} \in S_1 \cap M_1. \quad (128)$$

The first term shows the effect of deviations of firing rates from $1/2$, that is, the KL-divergence from $\hat{p}$ to p_0 . The second represents the effect of interaction shown by the KL-divergence from p to the independent model $\hat{p}$.

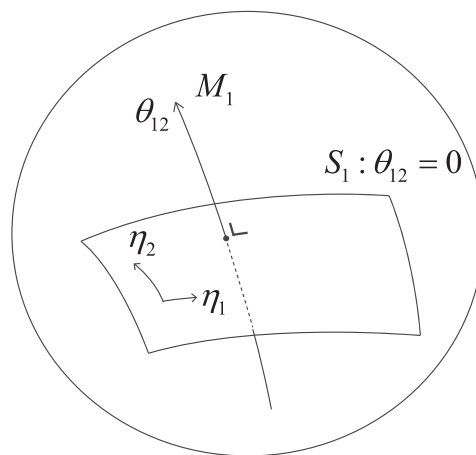


Figure 5. Orthogonal decomposition.

We next consider the case of $n = 3$ neurons. It has a hierarchical structure $S_1 \subset S_2 \subset S_3$. The S_1 is an independent model,

$$S_1 = \{p(\mathbf{x}) | p(\mathbf{x}) = p_1(x_1)p_2(x_2)p_3(x)\} \quad (129)$$

and η_i s show the firing rates of neurons. Since directions of θ^{ij} , θ^{123} are orthogonal to the directions of η_i , they together represent degrees of neural interactions orthogonal to the firing rates. Further, consider submanifold specified by $M_2 = \{\eta | \eta_i = c_i, \eta_{ij} = c_{ij}\}$, for fixed c_i and c_{ij} . The pair (η_i, η_{ij}) represent both the firing rates and pairwise joint firing rates, but do not specify the triple firing rate, $\eta_{123} = E[x_1x_2x_3]$. The quantity that is orthogonal to both the firing rates and pairwise firing rates is given by θ^{123} . So,

$$\theta^{123} = \log \frac{p_{111}p_{100}p_{010}p_{001}}{p_{110}p_{101}p_{011}p_{000}} \quad (130)$$

represents the triplewise interaction orthogonal to the firing and pairwise firing rates.

A distribution p is m -projected to S_2 and S_1 , where the KL-divergence is decomposed as

$$D_{KL}[p:p_0] = D_{KL}[p:\hat{p}_2] + D_{KL}[\hat{p}_2:\hat{p}_1] + D_{KL}[\hat{p}_1:\hat{p}_0] \quad (131)$$

by the Pythagorean theorem.

In a general model of n neurons, we consider the mixed coordinates of the k -cut

$$\xi_k = (\eta_i, \dots, \eta_{i_1 \dots i_k}; \theta^{i_1 \dots i_k + 1}, \dots, \theta^{1 \dots n}). \quad (132)$$

H_k represent the joint firing rates up to order k and $\Theta^k = (\theta^{i_1 \dots i_k + 1}, \dots, \theta^{1 \dots n})$ represents degrees of mutual interactions of orders higher than k orthogonal to H_k . Therefore, $\theta^{i_1 \dots i_k}$ represent the degrees of $(k+1)$ -th order interactions. The KL-divergence from p to the origin is decomposed as

$$D_{KL}[p:p_0] = \sum_{k=0}^{n-1} D_{KL}[\hat{p}_{k+1}:\hat{p}_k], \quad (133)$$

where $\hat{p}_n = p$, $\hat{p}_k$ is the projection of p to S_k and $\hat{p}_0 = p_0$. This shows that $D_{KL}[\hat{p}_{k+1}:\hat{p}_k]$ represents the effect of the order k interactions. We show only binary models, but we can generalise the theory to any hierarchical dually flat models.

6 Conclusions

Since Rao's proposal three quarters of a century ago, information geometry, the geometrical theory of a manifold of probability distributions has been developed widely, giving useful tools for various fields related to probability. This paper reviews only some of interesting structures to which information geometry gives remarkable contributions, such as dually flat structures and discusses some applications.

There are some new developments in this area. Wong (2018) gives a new mathematical idea, which generalises the Legendre transformation, and presents a new theory applicable to

projectively flat manifolds. The Pythagorean and projection theorems hold, where the Rényi divergence plays the role of the canonical divergence in a dually projectively flat manifold.

The Wasserstein distance gives another type of divergence in a manifold of probability distributions and has broad applicability. The distance is responsible for the metric structure of the base space on which distributions are defined. However, it is not invariant. It is interesting to find a theory connecting them (Amari *et al.*, 2019). Li & Zhou (2019) proposed a fundamental theory unifying the two geometries. See Amari & Matsuda (2021) for more detail on Wasserstein statistics. Amari (2016) discusses various applications.

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[Received October 2020; accepted June 2021]